

Proposed role of ATP in protein breakdown: Conjugation of proteins with multiple chains of the polypeptide of ATP-dependent proteolysis

(protein/turnover/energy dependence/isopeptide linkage)

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ABSTRACT The heat-stable polypeptide ATP-dependent proteolysis factor 1 (APF-1) of the reticulocyte proteolytic system forms covalent compounds with proteins in an ATP-requiring reaction. APF-1 and lysozyme, a good substrate for ATP-dependent proteolysis, form multiple conjugates, as was shown by comigration of label from each upon gel electrophoresis. Multiple bands were also seen with other substrates of the ATP-dependent proteolytic system, such as globin or α -lactalbumin. Analysis of the ratio of APF-1 to lysozyme radioactivities and of the molecular weights of the bands indicated that they consist of increasing numbers of the APF-1 polypeptide bound to one molecule of lysozyme. The covalent linkage is probably of an isopeptide nature, because it is stable to hydroxylamine and alkali, and polylysine is able to give conjugates of APF-1. Removal of ATP after formation of the ^{125}I -labeled APF-1 conjugates with endogenous proteins caused the regeneration of APF-1, indicating presence of an amidase. This reaction is thought to compete with proteases that may act on APF-1-protein conjugates, especially those containing several APF-1 ligands. A sequence of reactions in which the linkage of APF-1 to the substrate is followed by the proteolytic breakdown of the substrate is proposed to explain the role of ATP.

Since the discovery of the energy requirement of protein breakdown by Simpson in 1953 (1), this feature has been found to be universal in a variety of biological systems (for review, see ref. 2). Only recently have cell-free systems in which protein breakdown is stimulated by ATP been delineated (3-6).

We have resolved the ATP-dependent proteolytic system from reticulocytes into several components (7, 8). In the preceding paper (9) it was found that the heat-stable polypeptide of this system, ATP-dependent proteolysis factor 1 (APF-1), is bound to reticulocyte proteins in an ATP-specific reaction. We now show that this is due to the formation of a covalent, probably amide, linkage between the protein substrate and one or more APF-1 molecules. The role of this reaction in protein breakdown is considered.

METHODS

Preparation of Radiolabeled Proteins and Enzyme Fractions. The purification and radiolabeling of APF-1 from rabbit reticulocytes were described in the previous article (9). Crystalline hen egg white lysozyme (Worthington) and purified bovine α -lactalbumin (a generous gift of W. Klee) were labeled with Na^{125}I by a method similar to that described for the radioiodination of APF-1 (9). When used in unlabeled form these proteins were treated in the same way without isotope. Globin labeled with $[^3\text{H}]$ leucine was prepared as described (7). Fraction II (0.5 M KCl eluate from DEAE-cellulose) was prepared from

ATP-depleted rabbit reticulocytes (7). Poly(L-lysine) was obtained from Miles-Yeda (M_r 1500-8000).

Determination of the Formation of Compounds of ^{125}I -Labeled APF-1 with Protease Substrates. The reaction mixture contained, in a final volume of 20 μl : 100 mM Tris-HCl (pH 7.6), 3 mM dithiothreitol, 5 mM MgCl_2 , 2 mM ATP, 65 μg of protein of fraction II, 3 μg of ^{125}I -APF-1 ($0.2\text{--}1 \times 10^5$ cpm), and added substrate protein where indicated. After incubation at 37°C for 20 min, the reaction was terminated by boiling for 2 min in the presence of 25 mM Tris-HCl (pH 6.8), 5% (vol/vol) glycerol, 0.5% NaDodSO₄, and 0.5% 2-mercaptoethanol. The samples were electrophoresed on NaDodSO₄/polyacrylamide slab gels (13 $\times$ 13 $\times$ 0.12 cm) with the system of Laemmli (10). Unless otherwise stated, 10-14% gradients of acrylamide were used. The gels were stained, destained, dried, and autoradiographed as described (9). The autoradiograms were scanned with an EC-910 densitometer (E-C Apparatus, St. Petersburg, FL).

RESULTS

Formation of Covalent Compounds Between APF-1 and Protease Substrates in an ATP-Dependent Reaction. It was shown previously that labeled APF-1 becomes covalently bound to reticulocyte proteins in an ATP-dependent reaction (9). On autoradiograms of NaDodSO₄/polyacrylamide gels numerous radioactive protein bands could be detected, and it appeared reasonable to assume that these represented conjugates with endogenous protein substrates of proteolysis and not modifications of specific enzyme(s). In order to test this notion directly, ^{125}I -APF-1 was incubated with crude reticulocyte fraction II in the presence of ATP and unlabeled lysozyme [which is a substrate for the ATP-dependent proteolytic system (Table 1)]. As shown in Fig. 1, when increasing concentrations of unlabeled lysozyme were added, the label became associated with several bands of higher molecular weight than APF-1, which itself ran to the end of the gel. Some of these were at positions not labeled in the absence of lysozyme (compare tracks 3-5 to track 2). When ^{125}I -labeled lysozyme was incubated with unlabeled APF-1 in the presence of ATP, these same new bands were labeled (track 7, Fig. 1). This pattern also depended on the addition of ATP (Fig. 1, track 6) and APF-1 (not shown). The number of distinct bands containing ^{125}I -APF-1 or ^{125}I -lysozyme radioactivities varied from 5 to 7 in different experiments. Bands C₂-C₅ were most intensively labeled, whereas C₁, C₆, and C₇ contained much less radioactive material.

Fig. 2 shows similar findings with α -lactalbumin and globin,

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Abbreviation: APF-1, ATP-dependent proteolysis factor 1.

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Table 1. Substrates of the ATP-dependent proteolytic system

Substrate	Extent of proteolysis, %	
	-ATP	+ATP (2 mM)
Experiment 1		
^{125}I -Lysozyme (1 μg)	3.0	31.8
^3H]globin (10 μg)	1.7	5.4
Experiment 2		
^{125}I - α -Lactalbumin (1 μg)	12.5	48.8

Protein breakdown was assayed as described (7), except that the volume of the incubation mixture was 50 μl . The various labeled substrates were added at the amounts indicated and were incubated at 37°C for 1 hr in the presence of 130 μg of fraction II and 2 μg of APF-1.

which are also substrates of the ATP-dependent proteolytic system (Table 1). The fact that the heavier bands appear in the same position in each case may be related to their similar polypeptide weights of 16,000, 14,500, and 14,000 for globin, lysozyme, and lactalbumin, respectively. Only band C_2 was not seen in the control. The regular spacing of the new bands is noteworthy.

Characterization of the APF-Lysozyme Adducts. The most plausible explanation of the above findings is that in the ATP-dependent reaction several APF-1 molecules make covalent bonds with one substrate molecule. If several substrate molecules were involved in making these derivatives one might expect to see a very irregular distribution of the bands reflecting the heterogeneity of the protein substrates present.

The relative amounts of APF-1 and lysozyme radioactivities, in the various compounds were estimated by densitometry of

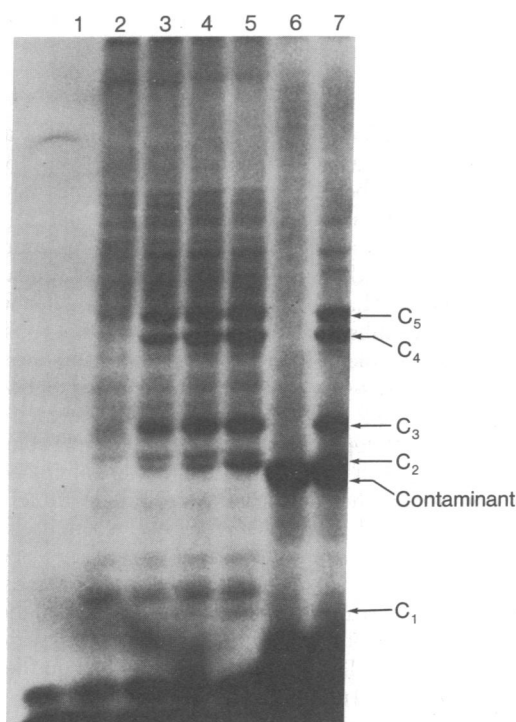


FIG. 1. Formation of covalent compounds between APF-1 and lysozyme in an ATP-dependent reaction. Tracks 1-5: compound formation with ^{125}I -APF-1. Track 1, without ATP; track 2, with ATP; tracks 3, 4, and 5, with ATP and 5, 10, or 25 μg of unlabeled lysozyme, respectively. Tracks 6 and 7: compound formation with ^{125}I -lysozyme. Conditions were as detailed in *Methods*, except that 5 μg of ^{125}I -lysozyme (40,000 cpm) and 3 μg of unlabeled APF-1 were used. Track 6, ATP omitted; track 7, with 2 mM ATP. Contamination in ^{125}I -labeled lysozyme is indicated.

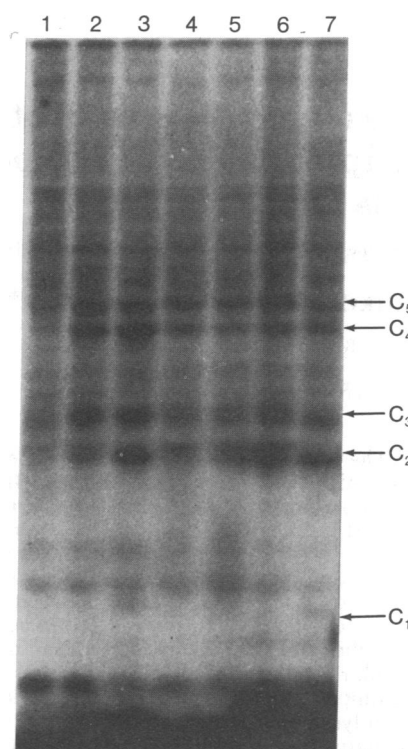


FIG. 2. Covalent linkage of APF-1 to various protease substrates. ^{125}I -APF-1 was incubated with fraction II and ATP. Track 1, without added substrate; tracks 2 and 3, with 10 or 50 μg of unlabeled lysozyme, respectively; tracks 4 and 5, with 10 or 50 μg of unlabeled α -lactalbumin, respectively; tracks 6 and 7, with 10 or 50 μg of unlabeled globin.

the autoradiograms (Fig. 3). These measurements were limited to C_3 - C_5 because the preparation of ^{125}I -lysozyme contained a contaminant in the region of C_2 , and C_1 , C_6 , and C_7 were too small for quantitative estimation. The accuracy of these estimates was limited by the necessary subtraction of a high background of endogenous ^{125}I -APF-1 conjugates present in this region (Fig. 3A, broken line). Nevertheless, a clear tendency

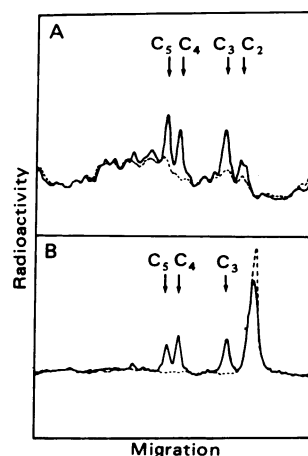


FIG. 3. Estimation of the relative amounts of ^{125}I -APF-1 and ^{125}I -lysozyme in the compounds. (A) Densitometric scan of ^{125}I -APF-1 (3 μg) incubated with unlabeled lysozyme (5 μg); the control (---) was incubated with ATP but without lysozyme. (B) Scan of ^{125}I -lysozyme (5 μg) incubated with unlabeled APF-1 (3 μg); the control (---) was a parallel sample incubated without ATP. The areas of the peaks above the respective controls were estimated by cutting them out and weighing them.

Table 2. Estimation of relative radioactivities and molecular weights of APF-1-lysozyme compounds

Compound	Radioactivity relative to C ₃ *			Molecular weight	
	¹²⁵ I-APF-1 (A)	¹²⁵ I-lysozyme (B)	Ratio (A/B)	Observed†	Calculated‡
C ₁				21,000	22,500
C ₂				34,000	30,500
C ₃	1.00	1.00	1.00	38,000	38,500
C ₄	1.11	0.87	1.28	47,000	46,500
C ₅	1.11	0.63	1.76	50,000	54,500

* Data from Fig. 3. Areas under the peaks were measured and calculated relative to the area of C₃.

† Data from Fig. 4.

‡ Assuming 1, 2, . . . 5 molecules of APF-1 per 1 undegraded molecule of lysozyme. The molecular weight of lysozyme is 14,500 and of APF-1 is about 8000, according to amino acid analysis (unpublished results).

for the increase in the ratio of ¹²⁵I-APF-1 to ¹²⁵I-lysozyme could be observed, because the amount of ¹²⁵I-APF-1 in compounds C₃–C₅ increased slightly, while that of ¹²⁵I-lysozyme decreased concomitantly. The relative radioactivities of ¹²⁵I-APF-1 and ¹²⁵I-lysozyme and the APF-1-to-lysozyme ratios in compounds C₃–C₅ are summarized in Table 2. The closest estimate that fits the radioactivity ratios in compounds C₃, C₄, and C₅ is 3, 4, and 5 molecules of APF-1 per molecule of lysozyme (the expected ratios are 1.33 and 1.67 for C₄ and C₅, respectively).

The molecular weights of the various compounds were estimated by NaDodSO₄/polyacrylamide gel electrophoresis (Fig. 4), and the observed values are summarized in Table 2. Table 2 also lists the calculated molecular weights of the various compounds, assuming that C₁–C₅ contain 1–5 molecules of APF-1 and 1 undegraded molecule of lysozyme. The observed molecular weights for C₁, C₃, and C₄ are in good agreement with the calculated values, but there are differences of 4000–5000 in the cases of C₂ and C₅, possibly indicating the loss of a fragment of this size by proteolysis.

Nature of the Covalent Bond. It was previously observed that the bonds between ¹²⁵I-APF-1 and endogenous proteins are stable to denaturing agents or to prolonged heating in the

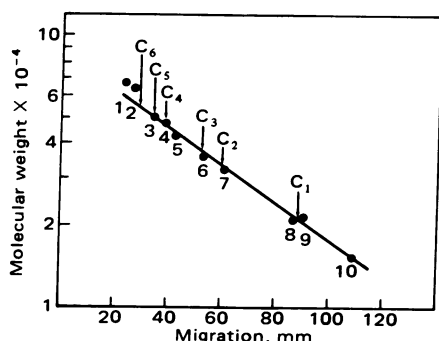


FIG. 4. Estimation of the size of the APF-1-lysozyme compounds by NaDodSO₄/polyacrylamide gel electrophoresis. Samples containing the compounds labeled with either ¹²⁵I-APF-1 or ¹²⁵I-lysozyme (see Fig. 1) were separated on a 12.5% acrylamide gel, together with the following marker proteins: 1, bovine serum albumin (*M_r* 68,000); 2, tetramer of globin, crosslinked (*M_r* 64,000); 3, glutamate dehydrogenase (*M_r* 50,000); 4, trimer of globin, crosslinked (*M_r* 48,000); 5, ovalbumin (*M_r* 43,000); 6, glyceraldehyde-3-phosphate dehydrogenase (*M_r* 36,000); 7, dimer of globin, crosslinked (*M_r* 32,000); 8, adenylate kinase (*M_r* 21,000); 9, soybean trypsin inhibitor (*M_r* 21,500); 10, globin monomer, (*M_r* 16,000).

Table 3. Covalent APF-1 adducts with polylysine

Additions	¹²⁵ I-APF-1 retained on CM-cellulose, %
None	1.0
ATP (2 mM)	2.4
Polylysine (1 mg/ml)	3.2
ATP + polylysine	10.0

¹²⁵I-APF-1 (3 μ g, 2.6×10^4 cpm) was incubated with fraction II (123 μ g of protein) and the above additions in an incomplete assay system (–ATP) for 15 min. The reaction mixture was applied to a CM-cellulose column (0.6 $\times$ 1.4 cm) equilibrated with 0.3 M potassium phosphate (pH 7.6) and washed with 6 column volumes of the same buffer. Radioactivity remaining on the column was determined with a Beckman Biogamma II counter.

presence of high concentrations of mercaptoethanol (9). According to the assay based on molecular size of derivatives of ¹²⁵I-APF-1, they were stable even during incubation at 37°C for 20 hr with 0.1 M NaOH or with 1 M hydroxylamine at pH 9.0. This suggests a peptide or amide linkage. The observation that several APF-1 molecules are attached to a single protein rules out the amino- or carboxy-terminal groups of the substrate protein as points of attachment. An amide linkage suggests that the ϵ -NH₂ group of lysine is involved in the bond. Polylysine was tested as a substrate and was shown to yield unique radioactive products that were tightly bound to CM-cellulose (Table 3).

Dynamic State of APF-1-Protein Derivatives. When ATP was removed with hexokinase plus deoxyglucose after the APF-1 compounds had formed, they decreased and eventually disappeared as electrophoretic bands as well as in the G-75

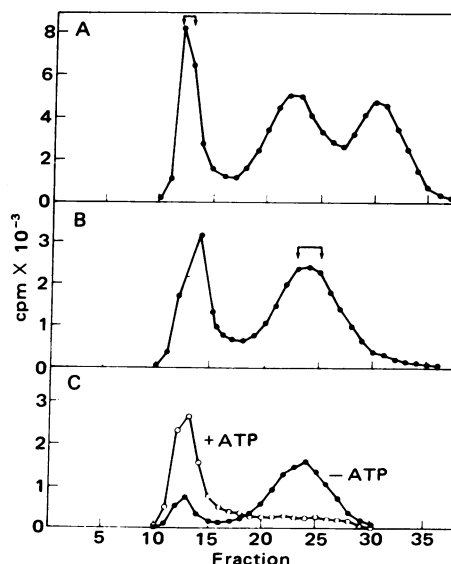


FIG. 5. Reutilization of free APF-1 released from compounds with substrates. (A) ¹²⁵I-APF-1 (12.8 μ g, 5.6×10^6 cpm) was incubated with fraction II (520 μ g) and ATP for 30 min, quenched with 0.1 M NaOH, and separated on a column (0.8 $\times$ 30 cm) of Sephadex G-75 equilibrated with 20 mM Tris-HCl (pH 7.6) and 1 mg of bovine serum albumin per ml. (B) The peak fractions from the void volume of A (marked by arrows) were collected and incubated at 37°C for 2 hr in a volume of 1.8 ml with 6 mg of fraction II in a reaction mixture lacking ATP, and were separated on a similar column. (C) The peak fractions of free APF-1 from B (marked by arrows) were collected and incubated at 37°C for 30 min in a volume of 1.0 ml with 3 mg of fraction II in the presence or absence of ATP, and separated on similar columns.

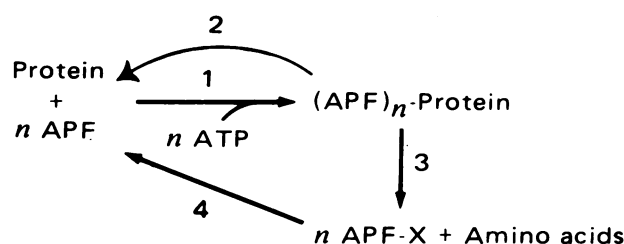


FIG. 6. Proposed sequence of events in ATP-dependent protein breakdown. See the text. 1, APF-1-protein amide synthetase (acting on lysine ϵ -NH₂ groups). 2, Amidase that allows correction when $n = 1$ or 2. 3, Peptidases that act strongly on (APF-1)_n derivatives, when $n > 1$ or 2. 4, Amidase for APF-1-X; X is lysine or a small peptide.

separation reported earlier (9). Fig. 5 shows that in this process APF-1 is regenerated and can be converted to high molecular weight derivatives again when incubated with ATP and fraction II.

DISCUSSION

By use of gel chromatography and NaDodSO₄ electrophoresis at least two enzymatic activities have been demonstrated here. The first is responsible for synthesis of a chemical bond between APF-1 and the ϵ -NH₂ group of lysine in polypeptides and proteins. Judging by its stability, the new bond may be an amide linkage. This activity, "APF-1-protein amide synthetase," is either precessive or has a high preference for free proteins that have APF-1 attached. This is shown by the appearance of derivatives with more than one APF-1 molecule under conditions of excess free substrate protein. The amino acid residue of APF-1 that contributes to the bond is not yet known. ATP is specifically required. A second enzyme activity leads to regeneration of APF-1 from APF-1-protein. This enzyme may be termed an amidase. The amidase may be acting on the APF-1-protein itself at the isopeptide bond, in which case it may be considered a "correction enzyme." Otherwise, or perhaps in addition, the amidase may be acting on degradation products of the APF-1-protein derivatives, if indeed these are the substrates for the proteolytic activity that is also observed in fraction II. In this case the amidase is essential if APF-1 is to play a catalytic role as a coenzyme of protein degradation. The amidase does not require ATP. As mentioned previously (9), the presence of APF-1-protein conjugates in the intact reticulocytes could explain the effect of ATP depletion of the cell in demonstrating the APF-1 requirement for protein breakdown.

Evidence that APF-1-proteins are intermediates in the breakdown of denatured protein as proposed in Fig. 6 is indirect and inconclusive at this time. The requirement for ATP spe-

cifically (3, 8) and APF-1 (7) for protein breakdown are conveniently explained if the adducts are intermediates. The precessive character of the first reaction of the model might be important in distinguishing between denatured proteins and native proteins with an exposed lysine. The complex containing enzyme, APF-1, and protein would be less likely to add a second APF-1 unit without first dissociating if conformational mobility of the protein limits access to a second lysine. The amidase would provide correction to adducts with few APF-1 ligands at a favored rate compared with proteolysis, which would depend on specific proteases that could recognize the APF-1 component and were especially directed to molecules with several APF-1 molecules attached. Protein breakdown occurring on a multi-enzyme complex has been an attractive hypothesis because of the general failure to detect small peptides as intermediates (11). Fraction II has recently been found (unpublished data) to contain at least four separable proteins that together with APF-1 are necessary for the ATP-dependent formation of trichloroacetic acid-soluble products of protein breakdown. Isolation studies should demonstrate whether these fractions participate in the proposed scheme.

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